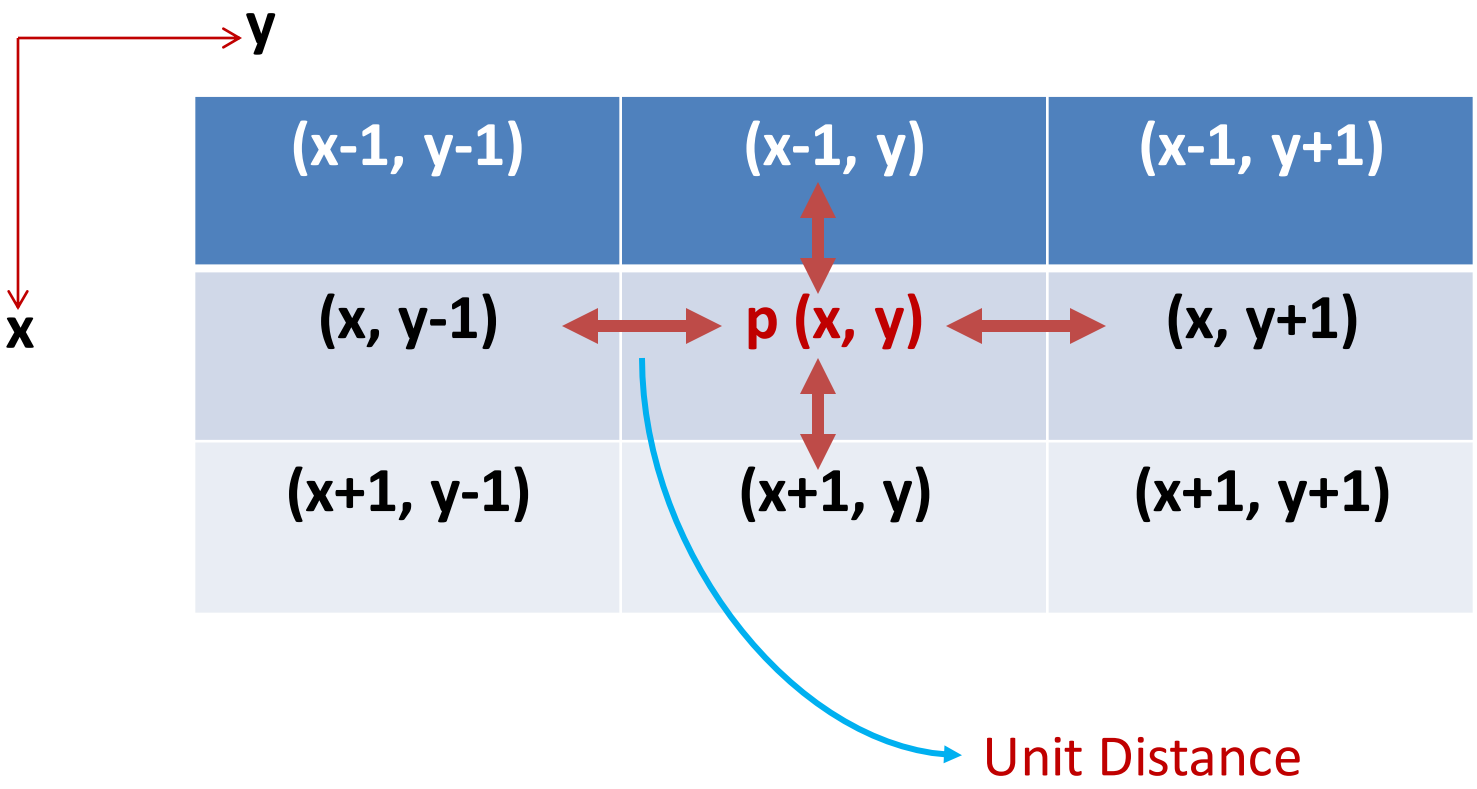
Pixel Geometry

Basic Relationships Between Pixels

- Neighborhood
- Adjacency
- Paths
- Connectivity
- Regions
- Boundary / Border / Contour
- Edge
- Distance Measures

Neighbors of a Pixel p at (x, y)



Neighbors of a Pixel p at (x, y)

- **4-neighbors of p**, denoted by $\mathbf{N_4(p)}$:
  (x-1, y), (x+1, y), (x,y-1), and (x, y+1).

- **4-diagonal neighbors of p**, denoted by $\mathbf{N_D(p)}$:
  (x-1, y-1), (x+1, y+1), (x+1,y-1), and (x-1, y+1).

Note: Some of the neighbors of pixel p lie outside the digital image if (x, y) is on the border of the image.

Neighbors of a Pixel p at (x,y)

- **8 neighbors of pixel p** is the union of 4-neighbors of p and 4-diagonal neighbors of p.

- It can be denoted as:

$$N_8(p) = N_4(p) \cup N_D(p)$$

Note: Some of the points of $N_4(p)$ and $N_D(p)$ fall outside the digital image if (x, y) is on the border of the image.

Adjacency

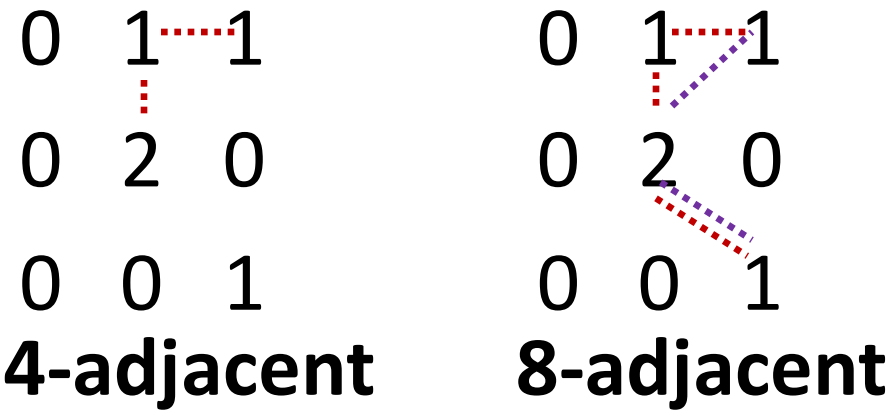
Let V be a set of intensity (gray-level) values used to define adjacency.

- **4-adjacency**: Two pixels p and q with values from V are 4-adjacent if q is in the set $N_4(p)$.

- **8-adjacency**: Two pixels p and q with values from V are 8-adjacent if q is in the set $N_8(p)$.

Examples: Adjacency

Suppose V = {1, 2}



4-adjacent **8-adjacent**

8-adjacency often generates ambiguities.

Adjacency

- **m-adjacency (mixed adjacency)**: Two pixels p and q with values from V are m-adjacent if

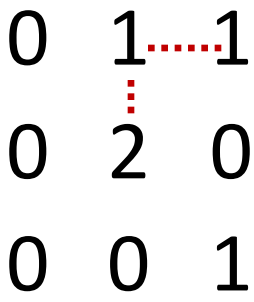
(i) q is in the set $N_4(p)$

or

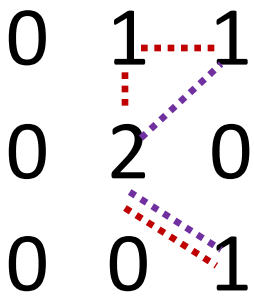
(ii) q is in the set $N_D(p)$ and the set $N_4(p) \cap N_4(q)$ has no pixels whose values are from V.

Examples: Adjacency

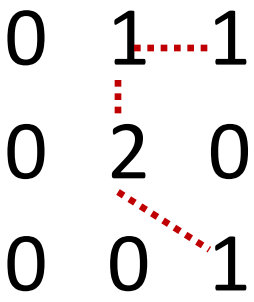
Suppose V = {1, 2}



4-adjacent



8-adjacent



m-adjacent

Mixed adjacency is a modification of 8-adjacency. It is introduced to eliminate the ambiguities that often arise when 8-adjacency is used.

Path

- A (digital) path (or curve) from pixel p with coordinates (x_0, y_0) to pixel q with coordinates (x_n, y_n) is a sequence of distinct pixels with coordinates

  $$(x_0, y_0), (x_1, y_1), \ldots, (x_n, y_n)$$

  where (x_{i-1}, y_{i-1}) and (x_i, y_i) are adjacent for $1 \leq i \leq n$.

- Here n is the length of the path.

Path

- If $(x_0, y_0) = (x_n, y_n)$, the path is closed path.
- We can define 4-, 8-, and m-paths based on the type of adjacency used.

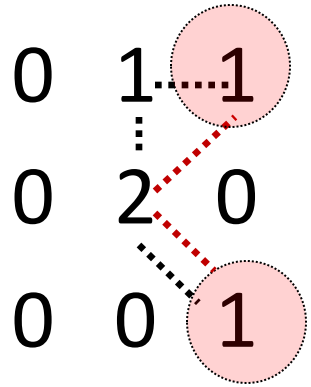
Examples: Path

Suppose V = {1, 2}

$0_{1,1}$ $1_{1,2}$ $1_{1,3}$

$0_{2,1}$ $2_{2,2}$ $0_{2,3}$

$0_{3,1}$ $0_{3,2}$ $1_{3,3}$



8-adjacent

The 8-path from (1,3) to (3,3):

(i) (1,3), (1,2), (2,2), (3,3)

(ii) (1,3), (2,2), (3,3)

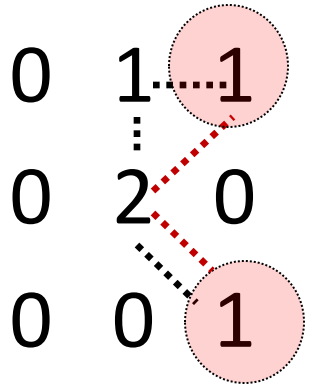
Examples: Path

Suppose V = {1, 2}

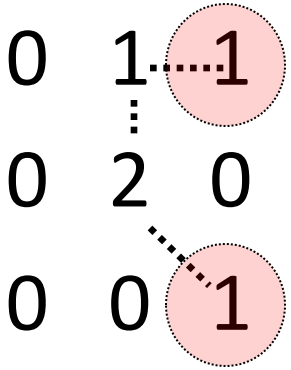
$0_{1,1}$ $1_{1,2}$ $1_{1,3}$

$0_{2,1}$ $2_{2,2}$ $0_{2,3}$

$0_{3,1}$ $0_{3,2}$ $1_{3,3}$



8-adjacent



m-adjacent

The m-path from (1,3) to (3,3):

(i) (1,3), (1,2), (2,2), (3,3)

Connectivity

- Let S represents a subset of pixels in an image. Two pixels p and q are said to be connected in S if there exists a path between them consisting entirely of pixels in S.

Connected Component

Let S represents a subset of pixels in an image.

- For every pixel p in S, the set of pixels in S that are connected to p is called a connected component of S.

- If S has only one connected component, then S is called connected set.

Region

- We call R a region of the image if R is a connected set.
- Two regions, R_i and R_j are said to be adjacent if their union forms a connected set.
- Regions that are not to be adjacent are said to be disjoint.

Boundary or Border

- The boundary of the region R is the set of pixels in the region that have one or more neighbors that are not in R.

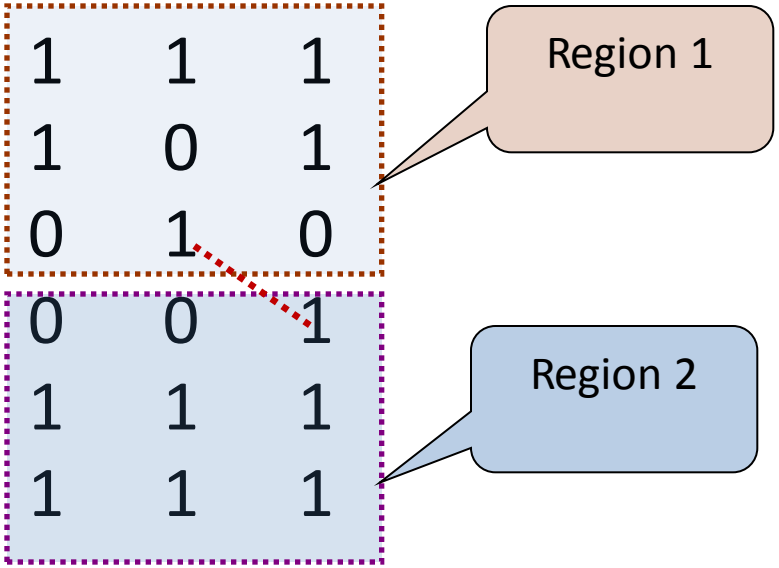
- If R happens to be an entire image, then its boundary is defined as the set of pixels in the first and last rows and columns of the image.

Foreground and Background

- An image contains K disjoint regions, R_l, l = 1, 2, …, K. Let R_u denotes the union of all the K regions, and let $(R_u)^c$ denote its complement.
- All the points in R_u is called foreground.
- All the points in $(R_u)^c$ is called background.

Question 1(a)

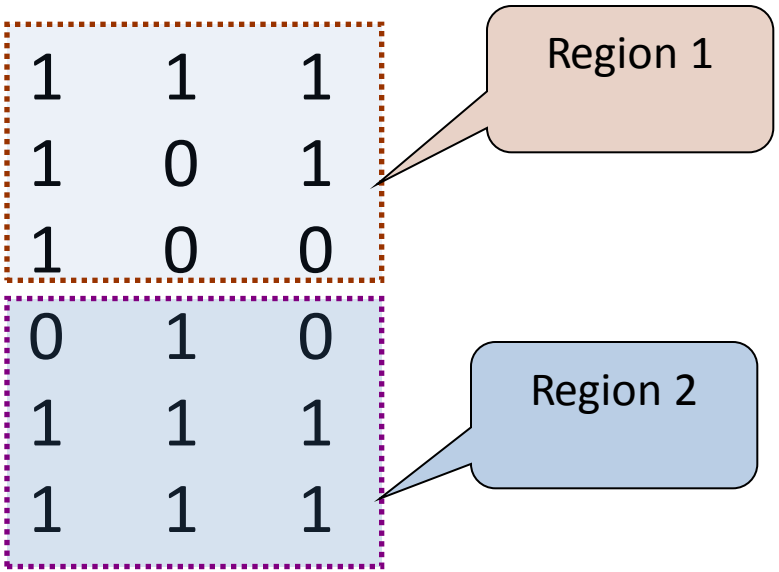
- **In the following arrangement of pixels, are the two regions (of 1s) 8-adjacent?**



Ans: YES

Question 1(b)

- **In the following arrangement of pixels, are the two regions (of 1s) 4-adjacent?**



Ans: NO

Question 2(a)

- **In the following arrangement of pixels, the circled point is part of the boundary of the 1-valued pixels if 4-adjacency is used, true or false?**

0	0	0	0	0
0	1	1	0	0
0	1	1	0	0
0	1	1	1	0
0	1	1	1	0
0	0	0	0	0

Ans: NO

Question 2(b)

- **In the following arrangement of pixels, the circled point is part of the boundary of the 1-valued pixels if 8-adjacency is used, true or false?**

0	0	0	0	0
0	1	1	0	0
0	1	1	0	0
0	1	1	1	0
0	1	1	1	0
0	0	0	0	0

Ans: YES

Edge and Boundary

- There is a key difference between these two concepts.
- The boundary of a finite region forms a closed path and thus a “global” concept.
- Edges are formed from pixels with derivative values that exceed a preset threshold.
- Thus the idea of an edge is a “local” concept that is based on a measure of gray level discontinuity at a point.

Edge and Boundary

- It is possible to link edge points into edge segments, and sometimes these segments are linked in such a way that correspond to boundaries, but this is not always the case.
- The one exception in which edges and boundaries correspond is in binary images.
- Depending on the type of connectivity and edge operators used, the edge extracted from a binary region will be the same as the region boundary.
- For the time being, it will be helpful to think of edges as intensity discontinuities and boundaries as closed paths.

Distance Measures

- Given pixels p, q and z with coordinates (x, y), (s, t), (u,v) respectively, the distance function D has following properties:

a. $D(p, q) \geq 0 \quad [D(p, q) = 0, \text{ iff } p = q]$

b. $D(p, q) = D(q, p)$

c. $D(p, z) \leq D(p, q) + D(q, z)$

Distance Measures

The following are the different Distance measures:

a. Euclidean Distance:

$$D_e(p, q) = [(x-s)^2 + (y-t)^2]^{1/2}$$

b. City Block Distance:

$$D_4(p, q) = |x-s| + |y-t|$$

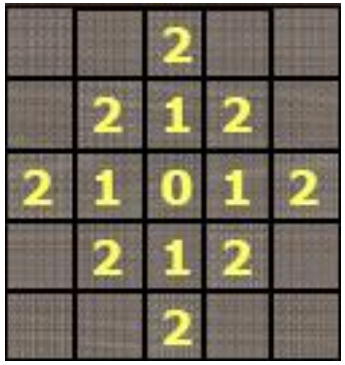
c. Chess Board Distance:

$$D_8(p, q) = \max(|x-s|, |y-t|)$$

Distance Measures

- For Euclidean distance measure, the pixels having a distance less than or equal to some value r from (x,y) are the points contained in a disk of radius r centered at (x,y).

- Pixels with D_4 distance from (x,y) form a diamond centered at (x,y).



Here r = 2

Distance Measures

- Pixels with D_8 distance from (x,y) form a square centered at (x,y).



Here r = 2

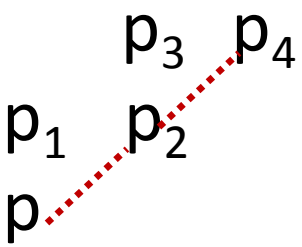
Distance Measures

- The D_4 and D_8 distances between p and q are independent of any paths that might exist between the points because these distances involve only the coordinates of the points.

- D_m distance between two pixels depends on the values of the pixels along the path, as well as the values of their neighbors.

Distance Measures

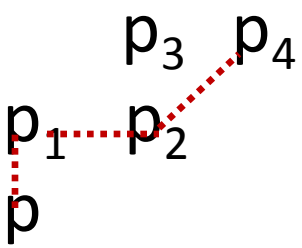
Consider the following arrangement of pixels and assume that p, p_2, and p_4 has value 1 and that p_1 and p_3 can have a value of 0 or 1.



Case I: V = {1}, p_1=0, p_3=0, then $D_m(p, p_4)$ = ? 2

Distance Measures

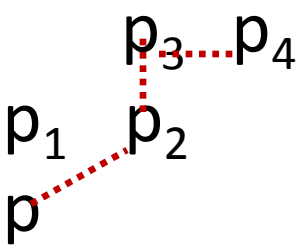
Consider the following arrangement of pixels and assume that p, p_2, and p_4 has value 1 and that p_1 and p_3 can have a value of 0 or 1.



Case II: V = {1}, p_1=1, p_3=0, then $D_m(p, p_4)$ = ? 3

Distance Measures

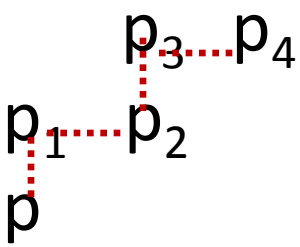
Consider the following arrangement of pixels and assume that p, p_2, and p_4 has value 1 and that p_1 and p_3 can have a value of 0 or 1.



Case III: V = {1}, p_1=0, p_3=1, then $D_m(p, p_4)$ = ? 3

Distance Measures

Consider the following arrangement of pixels and assume that p, p_2, and p_4 has value 1 and that p_1 and p_3 can have a value of 0 or 1.



Case IV: V = {1}, p_1=1, p_3=1, then $D_m(p, p_4)$ = ? 4

Linear and Nonlinear Operations

- Let H be an operator whose input and output are images, H is said to be a linear operator if, for any two images f and g and any two scalars a and b,

$$H(af + bg) = aH(f) + bH(g)$$

- The results of applying a linear operator to the sum of two images (that have been multiplied by the constants) is identical to applying the operator to the images individually, multiplying the results by the appropriate constants, and then adding those results.

Linear and Nonlinear Operations

Examples:

- An operator whose function is to compute the sum of K images.

- An operator that fails the test of the below equation is by definition nonlinear.

$$H(af + bg) = aH(f) + bH(g)$$

Examples:

- An operator that computes the absolute value of the difference of two images.